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Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection – Campus Placement & Internship System

Architecture Selection

We chose Layered Architecture for the Campus Placement & Internship System.

Architectural Choice

The system is divided into Presentation, Business, and Data/Integration layers. The Presentation Layer contains the User Interface (Student & Officer Views) Component. The Business Layer contains the Identity & Access Manager Component and Recruitment Drive Manager Component. The Data/Integration Layer contains the Pipeline Application & User Database Component.

Reason 1

Layered architecture provides a clear separation of responsibilities. User authentication, recruitment drive management, and profile/application data storage are handled by separate components. This makes the placement system easier to understand, maintain, test, and modify.

Reason 2

The architecture is suitable for the Campus Placement & Internship System because the system has closely related functions, such as login, recruitment drives, and applications, that can communicate through well-defined interfaces. It provides a simple and organized structure without the additional operational complexity of a microservices architecture.

Security Advantage

The layered design prevents the User Interface from directly accessing the user database. Authentication and authorization are handled by the Identity & Access Manager at the application/service boundary before requests reach the database, providing better control over sensitive student and officer data.

Performance Benefit

The separation of responsibilities allows common operations such as login verification and recruitment drive retrieval to be processed through the Identity & Access Manager and Recruitment Drive Manager, supported directly by the pipeline database. This avoids unnecessary processing at the user interface and reduces the network overhead that could occur with a distributed microservices design.

Component Modelling Diagram

Campus Placement & Internship System — Layered Architecture

- Presentation Layer
- Business Layer
- Data / Integration Layer

